Chapter 12, Problem 1RQ

Problem

What is the phase sequence of a three-phase motor for which $\mathbf{V}_{AN} = 220\underline{/-100^\circ}\text{ V}$ and $\mathbf{V}_{BN} = 220\underline{/140^\circ}\text{ V}$?

(a) *abc*

(b) *acb*

Step-by-step solution

Step 1 of 3

Refer to Figure 12.7 for the phase sequences in the textbook.

The phase sequence of a three-phase motor is,

$$\begin{aligned}\mathbf{V}_{AN} &= 220\angle -100^\circ\text{ V}\\ \mathbf{V}_{BN} &= 220\angle 140^\circ\text{ V}\\ &= 220\angle(-100^\circ + 240^\circ)\text{ V}\end{aligned}$$

Therefore,

$$\mathbf{V}_{CN} = 220\angle(-100^\circ + 120^\circ)\text{ V}$$

Step 2 of 3

For the *acb* or negative sequence, the phase equations are,

$$\begin{aligned}\mathbf{V}_{AN} &= 220\angle -100^\circ\text{ V}\\ \mathbf{V}_{BN} &= 220\angle(-100^\circ + 120^\circ)\text{ V}\\ &= 220\angle 20^\circ\text{ V}\end{aligned}$$

And,

$$\begin{aligned}\mathbf{V}_{CN} &= 220\angle(-100^\circ - 120^\circ)\text{ V}\\ &= 220\angle(-220^\circ)\text{ V}\end{aligned}$$

Therefore, option (b) is not the answer.

Step 3 of 3

For the *abc* or positive sequence, the phase equations are,

$$\begin{aligned}\mathbf{V}_{AN} &= 220\angle -100^\circ\text{ V}\\ \mathbf{V}_{BN} &= 220\angle 140^\circ\text{ V}\\ &= 220\angle(-100^\circ - 120^\circ)\text{ V}\\ &= 220\angle(-100^\circ + 240^\circ)\text{ V}\end{aligned}$$

Therefore,

$$\mathbf{V}_{CN} = 220\angle(-100^\circ + 120^\circ)\text{ V}$$

Therefore, the correct answer is option $\boxed{(\text{a})}$.